

- Q.26 Explain with diagrams, the formation of steam.
- Q.27 Define priming, foaming, and carryover in the context of boiler operation.
- Q.28 Differentiate between soft water, and hard water. Provide examples of applications for each type of water.
- Q.29 Discuss various boiler accessories such as ID fan, FD fans, and superheaters.
- Q.30 Explain any one type of steam trap used in Industries.
- Q.31 Write about the some important refractories with their Failure of refractories.
- Q.32 Difference between sludge and scale.
- Q.33 Briefly explain about insulation with its types.
- Q.34 Explain zeolite process with diagram.
- Q.35 Discuss about layout of piping used in industry.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Describe the construction and working principles of natural draft cooling tower, and their roles in dissipating heat from industrial processes.
- Q.37 What is boiler. Discuss Cochran boiler with diagram and its advantages.
- Q.38 Explain the concept of external treatment for water conditioning and discuss the processes involved such as Ion Exchange Process with diagram.

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Roll No.

6th Sem / Chem, P &P, Chem Engg. (Spl. Paint Tech.) Subject:- Process Plant Utilities / Proc. Utilities

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 What are the two main categories of water hardness?
- Acidic and alkaline water
 - Soft and hard water
 - Clear and cloudy water
 - Temporary and permanent hardness
- Q.2 In the h-s diagram (Mollier chart), What does the horizontal axis typically represent?
- temperature
 - Pressure
 - Enthalpy
 - Specific volume
- Q.3 Which type of refractory is known for its low thermal conductivity and is used for insulation purposes?
- Acid refractory
 - Basic refractory
 - Insulating refractory
 - Special refractory
- Q.4 Which of the following is a common cause of temporary hardness in water?
- Calcium sulfate
 - Calcium carbonate
 - magnesium chloride
 - Sodium chloride
- Q.5 What type of cooling tower relies on natural airflow for heat removal?

- a) Natural draft cooling tower
 - b) Forced draft cooling tower
 - c) Induced draft cooling tower
 - d) Closed-circuit cooling tower
- Q.6 What is steam primarily composed of?
- a) Oxygen b) Hydrogen
 - c) Water vapor d) Carbon dioxide
- Q.7 What is the main aim of insulation in industry?
- a) To generate heat content
 - b) To insulate against heat transfer
 - c) All of above
 - d) none of above
- Q.8 Which process involves the use of resins to remove ions from water through an exchange process?
- a) Lime Soda Process
 - b) Ion Exchange Process
 - c) Phosphate Conditioning
 - d) Calgon conditioning
- Q.9 Which refrigerant is commonly used in industrial refrigeration system due to its excellent thermodynamic properties and safety record?
- a) Ammonia b) carbon dioxide
 - c) Cryogeme d) Antifreeze
- Q.10 Which type of boiler is known for its water-tube design and high steam generation capacity?
- a) Babcock Wilcox boiler
 - b) Soot blower
 - c) Nestler boiler
 - d) none of above

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Name the type of lime soda process.
- Q.12 Name one type of steam ejector.
- Q.13 Name the primary classification of refractories based on their chemical properties?
- Q.14 What is dryness fraction?
- Q.15 Write one application of insulation?.
- Q.16 Write one use of pump.
- Q.17 Name any two boilers used in industry.
- Q.18 _____ is the term for the process of recharging resins in an ion exchange system?
- Q.19 What is full form of BOD.
- Q.20 Refrigeration stands for _____?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Define water hardness and provide insights into its primary cause. Differentiate between temporary and permanent hardness, providing examples of each.
- Q.22 What is “Boiler corrosion”, and why is it a concern in steam generation?
- Q.23 Explain the critical factors to consider when selecting refractories for specific industrial applications.
- Q.24 Write about selection of refrigerant.
- Q.25 Write about the overview of types of boilers, including Babcock Wilcox boiler with their unique features and applications.